Name: Logan A. Reed

Degree and Major: Bachelor Degree in Cybersecurity

Advisor Name: Prof. Julie Henderson

Expected Graduation Date: 2026

Problem Statement:

People going into Cybersecurity should have some level of experience building and securing networks. Oftentimes labs and projects in the classroom can't cover everything that goes into networks. This is important because if cybersecurity specialists knew how networks were built from the ground up, they can have an easier time finding problems within the network. Knowing how to build a network will help maintain and change it in the future as well.

My solution is having a situation where the student would have a practice range to build and secure a network to a certain standard. This will give the student experience and knowledge for the field. Without a big practice range to give the student experience within the field it will make real world problems more difficult, but giving them this practice range will help them maintain networks easier and longer.

Project Description:

My project will be creating and securing a medium sized corporate building's network using the free program CISCO Packet Tracer. The program is a good substitute for physical hardware devices by simulating them instead. I will document each step taken to secure the network and what doing that step will contribute to securing the network.

Software and Equipment Needed:

CISCO Packet Tracer is not a very complicated program and doesn't use a lot of processing power. It is free to use on any computer and can even run smoothly on laptops with less than 8GB of RAM.

Personal Motivation:

Doing this project will allow me to learn more about network creation and securing as well. It will also strengthen my skills in teaching others how to do so.

Outline of Future Research Efforts:

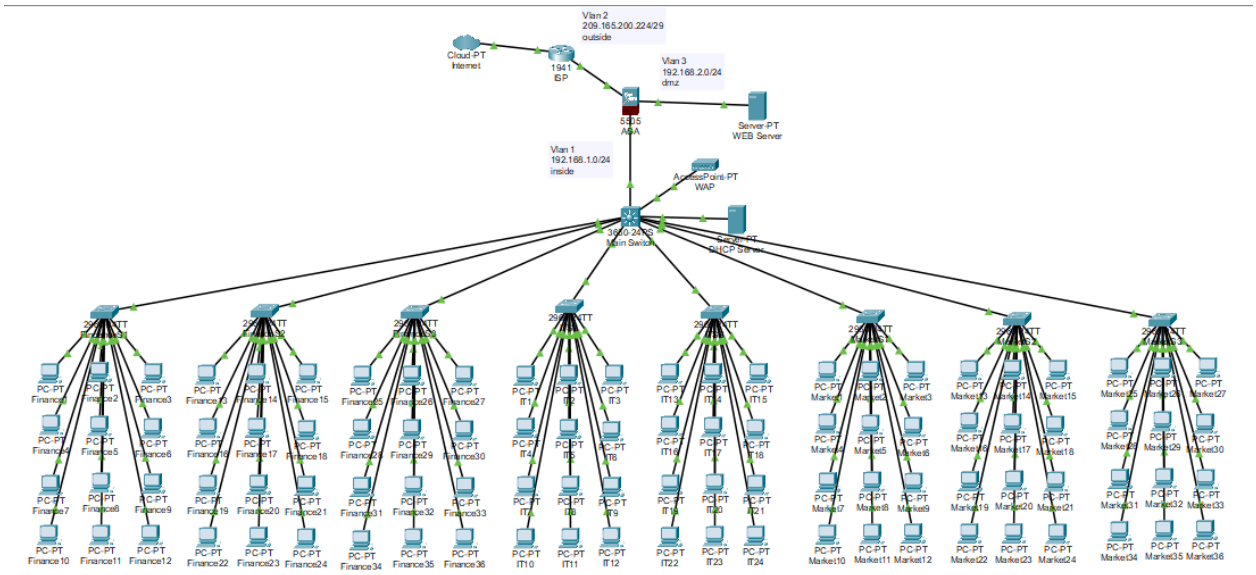
Using my own knowledge of networks and cybersecurity as well as researching techniques and configurations I will create and secure a full network for a medium sized corporation and then create a practice range based on it.

Schedule:

Date	**Task to have done by date**
10/3/25	**Set up Network ASA so they connect to the Outside Router and Main Switch. Document steps taken along the way.**
10/10/25	**Set up the main buildings DHCP/DNS, Access Wireless Point, and WEB Servers. Document steps taken along the way.**
10/17/25	**Set up the switches for Financing, IT, and Marketing subnets so they connect to the Main Switch. Document steps taken along the way.**
10/24/25	**Create scripts for setting up and hardening every computer in the Financing, IT, and Marketing subnets so they can be connected to the network. Document steps taken.**
10/31/25	**Create Security Plan using NIST Framework**
11/7/25	**Harden Main Building's devices Using Security Plan**
11/21/25	**Using the documentation taken, compile it all into a step by step lab so that people can copy the network. (Translate into spanish if available)**
11/24/25	**Create Test Plan and Finish Project**

🗓 Date 2/24/26	**Security Plan and Demo Ready**
🗓 Date 3/3/26	**Finish Student Survey and Start User Testing**
🗓 Date 3/24/26	**Finish Testing**
🗓 Date 3/31/26	**Have any issues found from testing fixed and start defense presentation**
🗓 Date 4/14/26	**Finish presentation and have all materials together in a github repo**

Network Diagram



Requirements:

- Teaches the student how to configure a Router for the company's network:
    - ID: 1
    - Type: Functional
    - Description: Using Router model "1941" the lab teaches the students how to set up a router by changing the default name and password on the router. The router should also have the used connections configured by setting IP addresses for them.

    - Rationale: This is one of the most important parts of any network because it is the gateway between the internet and the rest of the network
    - Criterion: The router should have packets able to go to and from the router.
    - Priority: 1
    - Dependencies: None
- Teaches the student how to set up a Network Firewall between the company's network and ISP:
    - ID: 2
    - Type: Security
    - Description: Set up a hardware firewall with the ASA model "5506-X" configuring policies on the device to prevent malicious packets from entering the network from the internet.
    - Rationale: Securing the network from the internet is an important step in hardening a network.
    - Criterion: The firewall should monitor and prevent malicious packets entering the network
    - Priority: 2
    - Dependencies: None
- Teaches the student how to set up the main switch for the main building and connecting it to the second building:
    - ID: 3
    - Type: Functional

    - Description: Using the Switch Model “3650-24PS” the student will learn how to configure a switch to connect the several other switches inside the network using trunking and straight through copper wires. The switch will need to have the IP addresses assigned for each used port. The student will add the “GLC-LH-SMD” module to the switch in order for a fiber optic wire to be used to connect the second building. The student will also learn how to connect the DHCP, DNS, and WEB servers to the switch using copper cross-over cables. The students will also connect the switch to the wireless access point firewall using another copper cross-over wire.
    - Rationale: This provides connectivity within the network along with organization and speed.
    - Criterion: The switch should be able to connect the other switches together allowing traffic between all the different subnets. It should also allow traffic to the 2nd building.
    - Priority: 3
    - Dependencies: 1
- Teaches the student how to set up three switches to connect many computers inside of the marketing subnet:
    - ID: 4
    - Type: Functional
    - Description: The lab teaches the student how to set up the model “2960” switches so that traffic can go to and from all the computers inside the marketing subnet / branch of the company.

    - Rationale: This allows traffic between the marketing subnet improving organization and efficiency
    - Criterion: The switches should allow any computer inside the subnet to receive and send packets to and from other authorized parts of the network.
    - Priority: 4
    - Dependencies: 3
- Teaches the student how to set up two switches to connect many computers inside of the IT subnet:
    - ID: 5
    - Type: Functional
    - Description: Teaches the student how to set up the model “2960” switches to allow for traffic inside the IT subnet / branch within the company’s network
    - Rationale: This improves organization and efficiency within the network along with connectivity.
    - Criterion: All computers within the IT department should be able to receive and send packets to and from the network and other computers in the network.
    - Priority: 5
    - Dependencies: 3
- Teaches the student how to set up three switches to connect many computers inside of the financing subnet:
    - ID: 6
    - Type: Functional

    - Description: Teaches the students how to set up the model “2960” switches to connect all the computers inside the financing subnet / branch to each other and other parts of the network.
    - Rationale: This improves organization and efficiency inside the subnet along with connectivity.
    - Criterion: All computers within the subnet should be able to receive and send packets to and from the rest of the network and each other.
    - Priority: 6
    - Dependencies: 3
- Teaches the students how to harden the switch connecting the other switches:
    - ID: 7
    - Type: Security
    - Description: The students will learn how to harden the “2960” switch connecting the other switches and subnets. The students will shut off unused ports and connections.
    - Rationale: This makes it so there are not as many ways for attackers to enter the network.
    - Criterion: Any unused ports should be turned off and new devices connecting to it should not work unless the connection is opened for them
    - Priority: 7
    - Dependencies: 3
- Teaches the students how to harden the switches connecting the marketing subnet:
    - ID: 8

    - Type: Security
    - Description: Teaches the student how to secure and protect the model “2960” switches that are used to connect the marketing subnet’s / branch’s computers together. The students will turn off unused ports and connections.
    - Rationale: This makes it so that the switches don't allow unauthorized access to the subnet and makes it more difficult for attackers to connect new devices.
    - Criterion: Any unused ports or connections should be shut off.
    - Priority: 8
    - Dependencies: 4
- Teaches the students how to harden the switches connecting the IT subnet:
    - ID: 9
    - Type: Security
    - Description: Teaches the student how to secure and protect the model “2960” switches that are used to connect the IT subnet’s / branch’s computers together. The students will turn off unused ports and connections.
    - Rationale: This makes it so that the switches don't allow unauthorized access to the subnet and makes it more difficult for attackers to connect new devices.
    - Criterion: Any unused ports and connections should be turned off
    - Priority: 9
    - Dependencies: 5
- Teaches the students how to harden the switches connecting the financing subnet:
    - ID: 10
    - Type: Security

    - Description: Teaches the student how to secure and protect the model “2960” switches that are used to connect the financing subnet’s / branch’s computers together. The students will turn off unused ports and connections.
    - Rationale: This makes it so that the switches don't allow unauthorized access to the subnet and makes it more difficult for attackers to connect new devices.
    - Criterion: Any unused ports and connections should be turned off.
    - Priority: 10
    - Dependencies: 6
- Teaches the students how to set up and configure the 24 computers in IT:
    - ID: 11
    - Type: Functional
    - Description: The students will learn how to set up 24 PC’s in the IT subnet by assigning IP addresses and subnet masks through the DHCP server and connecting them to the two switches in the subnet using copper cross-over wires.
    - Rationale: The computers would be used by the employees inside the company
    - Criterion: Every computer should be able to communicate with the rest of the network. An easy test is to use the broadcast command.
    - Priority: 11
    - Dependencies: 9
- Teaches the students how to set up the DHCP server inside the main building:
    - ID: 12
    - Type: Functional

    - Description: The students will learn how to set up a DHCP server for the main building's devices. The students will set up a subnet each for marketing, IT, financing, and wireless connections that get IP addresses automatically. They will connect the server to the main switch.
    - Rationale: The server makes it so that each computer does not need to be set up individually.
    - Criterion: Every device in the network with a DHCP IP address should have an address and any new devices connecting through the wireless access point should also get an IP address.
    - Priority: 12
    - Dependencies: 3
- Teaches the students how to harden and secure the 24 computers within the IT subnet:
    - ID: 13
    - Type: Security
    - Description: The students will learn how to secure all 24 computers within the IT subnet by creating group policies, making password requirements, and enabling firewalls.
    - Rationale: The computer security makes it so it is even harder to get into the network. By securing active computers it makes it so attackers can't use employee computers to penetrate the network.
    - Criterion: Every computer should have the IT subnets correct group policy, have the firewalls enabled, and have strong password requirements.
    - Priority: 13

    - Dependencies: 11
- Teaches the students how to set up the main building DNS server
    - ID: 14
    - Type: Functional
    - Description: The students will learn how to set up a DNS server and connect it to the main switch in the main building with a copper cross-over wire.
    - Rationale: The DNS server is an important part of a network allowing communication on the network.
    - Criterion: The DNS server should be able to be pinged by other computers in the network and be functioning.
    - Priority: 14
    - Dependencies: 3
- Teaches the students how to set up the WEB server
    - ID: 15
    - Type: Functional
    - Description: The students will learn how to set up the company's WEB server that runs the company's portal website.
    - Rationale: The WEB server is a big part of the functionality and organization of a company.
    - Criterion: Any other computer in the network should be able to ping the WEB server and connect to the website.
    - Priority: 15
    - Dependencies: 3

- Teaches the students how to secure the WEB server
    - ID: 16
    - Type: Security
    - Description: The student will learn how to harden the company WEB server by changing default passwords and names to stronger ones, disabling unused services, limiting entry points, and enabling firewalls.
    - Rationale: The WEB server is a big asset to the company and will likely be targeted during an attack. Securing it will help deter the attacker from attacking it.
    - Criterion: Outside devices should not be able to connect to the WEB server and the password should be very strong.
    - Priority: 16
    - Dependencies: 15
- Teaches the students how to set up the 36 computers in the Marketing subnet:
    - ID: 17
    - Type: Functional
    - Description: The students should learn how to setup and configure 36 computers inside the Marketing subnet by connecting them to the three switches inside the subnet with copper cross-through wires. The computers should also have the correct IP addresses assigned to them through the DHCP server.
    - Rationale: Employees in the marketing subnet would be working off of these computers.
    - Criterion: All the computers should be able to send and receive traffic from the rest of the network. A good test is using a broadcast command.

    - Priority: 17
    - Dependencies: 8
- Teaches the students how to setup the 36 computers inside the Financing subnet:
    - ID: 18
    - Type: Functional
    - Description: The students should learn how to set up all the computers inside the Financing subnet by connecting them to the three switches inside the subnet with copper cross-through wires. The computers should also have the correct IP addresses assigned to them through the DHCP server.
    - Rationale: Employees within the Financing branch will be working from these computers
    - Criterion: Every computer inside the Financing subnet should be able to send and receive traffic from the rest of the network. A good test is using the broadcast command.
    - Priority: 18
    - Dependencies: 10
- Teaches the students how to harden and secure the 36 computers within the Marketing subnet:
    - ID: 19
    - Type: Security
    - Description: The students will learn how to secure all 36 computers within the Marketing subnet by creating group policies, making password requirements, and enabling firewalls.

    - Rationale: The computer security makes it so it is even harder to get into the network. By securing active computers it makes it so attackers can't use employee computers to penetrate the network.
    - Criterion: Every computer should have the Marketing subnet's correct group policy, have the firewalls enabled, and have strong password requirements.
    - Priority: 19
    - Dependencies: 17
- Teaches the students how to harden and secure the 36 computers within the Financing subnet:
    - ID: 20
    - Type: Security
    - Description: The students will learn how to secure all 36 computers within the Financing subnet by creating group policies, making password requirements, and enabling firewalls.
    - Rationale: The computer security makes it so it is even harder to get into the network. By securing active computers it makes it so attackers can't use employee computers to penetrate the network.
    - Criterion: Every computer should have the Marketing subnet's correct group policy, have the firewalls enabled, and have strong password requirements.
    - Priority: 20
    - Dependencies: 18
- Teaches the students how to set up and configure the wireless access point in the main building:

    - ID: 22
    - Type: Functional
    - Description: The students will learn how to set up a WAP to allow wireless devices to connect to the network. The WAP will assign IP addresses using the DHCP server and should be connected to the firewall between it and the rest of the network with a copper cross-through wire. The WAP should also have a IP address itself.
    - Rationale: Employees will need to bring in devices like laptops and cell phones to work and will need to access the network through them. The WAP will allow this.
    - Criterion: All devices connecting through the WAP should be able to access the rest of the network.
    - Priority: 22
    - Dependencies: 21
- Teaches the students how to secure and harden the Wireless Access Point in the Main Building:
    - ID: 23
    - Type: Security
    - Description: The students will learn how to harden the wireless access point inside the main building by changing the default username and password, disabling unused services, and enabling the correct security and internal firewalls.
    - Rationale: Securing the WAP will prevent attackers from using it to gain entry into the network.

    - Criterion: Unused services should be turned off and the correct security and firewalls should be enabled
    - Priority: 23
    - Dependencies: 22
- There will be a lab to teach students how to setup the network
    - ID: 38
    - Type: Usability
    - Description: A lab that will teach the students how to set up the network step by step split up in sections and subsections.
    - Rationale: A step by step guide is required to teach students how to create and secure their network
    - Criterion: The lab should be comprehensible and simple enough for college students to learn how to build and secure a network. There should be sections and subsections splitting the lab into doable work loads. The end product of the students network should match the network built by myself. The lab should be educational and give reasons behind each step.
    - Priority: 38
    - Dependencies: 1 - 37
- Lab will have pictures for students to reference when going through steps:
    - ID: 39
    - Type: Usability
    - Description: There should be pictures to help students understand what they are looking for and how their network should look throughout the steps.

    - Rationale: Many people are visual learners so pictures can help with learning and keeping the students engaged with the lab.
    - Criterion: At least one picture at the end of each subsection of the lab to give the students a good idea of what has been done.
    - Priority: 39
    - Dependencies: 38
- Lab will have a spanish version:
    - ID: 40
    - Type: Cultural
    - Description: A full translation of the lab will be in spanish
    - Rationale: Many people have Spanish as their first language in America and people have a better time learning things in their native language.
    - Criterion: There should be a full translation of the lab that meets the same criterion as the original lab.
    - Priority: 40
    - Dependencies: 38
- The lab with use multiple colors to stylize it:
    - ID: 41
    - Type: Look and Feel
    - Description: The lab should have multiple colors to highlight certain areas or be easier to read.
    - Rationale: More colors makes the lab seem more professional and thought out and also can make some things easier to read or see.

    - Criterion: There should be at least 2 different colors used and at most 4 different colors used inside the lab
    - Priority: 41
    - Dependencies: 38

Future Enhancements

- Teaches the students how to set up the 2nd buildings main switch:
    - ID: 24
    - Type: Functional
    - Description: The students will learn how to connect the 2nd main switch to the main building's main switch with a fiber optic cable. The switch should also be connected to the two switches in the sales subnet with copper straight-through wires. There will also be connections to the 2nd building's DHCP and DNS servers using copper cross-over wires. The switch will also be connected to the firewall between itself and the wireless access point for the 2nd building with a copper cross-over cable. The main switch will have an IP address assigned to it as well.
    - Rationale: The purpose of the 2nd switch is to ease the load on the first switch and also to connect the two buildings so they are still in the same network.
    - Criterion: All devices should be connected with the correct wires and the switch should be correctly configured.
    - Priority: 24
    - Dependencies: 3
- Teaches the student how to secure and harden the 2nd building's main switch:

    - ID: 25
    - Type: Security
    - Description: The students will turn off all unused ports and connections in the switch and make strong credentials to the device.
    - Rationale: Since the switch is acting as a hub for the second building it needs to be secured well because if it isn't then an attacker can gain easy access to the network.
    - Criterion: Unused ports and connections should be turned off and strong credentials should be in effect.
    - Priority: 25
    - Dependencies: 24
- Teaches the students how to setup and configure the 2nd Building's DHCP server:
    - ID: 26
    - Type: Functional
    - Description: The students will learn how to set up a DHCP server to assign IP addresses to the sales subnet and devices connecting through the second building's wireless access point. The DHCP server will be connected to the 2nd building's main switch with a copper cross-over wire and will have a static IP address.
    - Rationale: Setting up a DHCP server can allow for easy expansion and organization inside the network.
    - Criterion: The DHCP server should be able to assign IP addresses to the sales subnet and the devices connecting to the 2nd building's wireless access point.

    - Priority: 26
    - Dependencies: 24
- Teaches the students how to harden the 2nd Building's DHCP server
    - ID: 27
    - Type: Security
    - Description: Students will learn how to secure the 2nd buildings DHCP server by enabling firewalls, creating strong credentials, disabling unnecessary services, and enabling the correct security.
    - Rationale: Hardening the DHCP server is a good thing because that is how computers in the network get their IP addresses.
    - Criterion: Unused services should be disabled, firewalls should be enabled, strong credentials should be used, and the correct security should be enabled.
    - Priority: 27
    - Dependencies: 26
- Teaches the students how to setup and configure the 2nd buildings DNS server:
    - ID: 28
    - Type: Functional
    - Description: Students will learn how to connect the DNS server in the 2nd building to the 2nd building main switch using a copper cross-over wire and assigning a static IP address to the server.
    - Rationale: Setting up the DNS server will allow outside domains to be contacted by receiving and sending traffic between them.

    - Criterion: Devices in the 2nd building should be able to connect to outside domains.
    - Priority: 28
    - Dependencies: 24
- Teaches the students how to harden the 2nd building's DNS server:
    - ID: 29
    - Type: security
    - Description: The students will learn how to harden the 2nd building's DNS server by disabling unused services, creating strong credentials, and enabling firewalls.
    - Rationale: Securing a DNS server is important for the security of the network because it is often connected to outside the network.
    - Criterion: Firewalls should be enabled, unused services should be disabled, and strong credentials should be being used.
    - Priority: 29
    - Dependencies: 28
- Teaches the students how to setup and configure the firewall between the 2nd building's wireless access point and the 2nd building's main switch
    - ID: 30
    - Type: Security
    - Description: The students will learn how to setup the firewall between the 2nd buildings wireless access point and its main switch by connecting the firewall to the switch and WAP with copper cross-over cables and assigning an IP address to

the firewall. ACL's, Firewall settings and strong credentials will be enabled as well.

    - Rationale: Putting a firewall between the WAP and Main switch is necessary since the firewall between the internet and network won't protect against the wireless devices trying to connect.
    - Criterion: Strong credentials should be used, ACL's should be configured, and firewall settings should be correct and the firewall should allow authorized traffic through and be connected to the rest of the network.
    - Priority: 30
    - Dependencies: 24
- Students will learn how to set up the 2nd building's wireless access point:
    - ID: 31
    - Type: Functional
    - Description: The students will set up the Wireless Access Point for the 2nd building by assigning it an IP address and connecting it the the firewall between itself and the 2nd building's main switch with a copper cross-over cable. It will allow for connection of wireless devices to the network through the 2nd Building. Using IP addresses from the DHCP server any devices connecting will be assigned IP addresses.
    - Rationale: Wireless connection to the main building's wireless access point won't be possible from the 2nd building and many employees will bring in their own devices that require connection to the network through wireless means.

    - Criterion: Any wireless device connecting to the network should have an IP address and be able to send and receive traffic throughout the network.
    - Priority: 31
    - Dependencies: 30
- Teaches the students how to secure and harden the 2nd building's wireless access point:
    - ID: 32
    - Type: Security
    - Description: The students will disable any unused services, create strong credentials for the device, enable firewalls, use the correct security and require a password to connect to the network.
    - Rationale: Wireless access points are often open to any other devices in the area so need a lot of security to make sure they aren't as much of a security threat.
    - Criterion: Firewalls should be enabled, strong credentials for the device should be used, correct security should be enabled, there should be a password for the wifi, and any unused services should be disabled.
    - Priority: 32
    - Dependencies: 31
- Teaches the students how to setup and configure the switches for the sales branch:
    - ID: 33
    - Type: Functional
    - Description: The students will set up switches for the sales subnet inside the 2nd building by connecting them to the 2nd building's main switch with copper straight-through cables and assigning static IP addresses to them. They should

also have connections to the computers within the subnet using copper cross-over cables.

    - Rationale: Switches improve organization and efficiency within the network and also allow for expansion and many connections.
    - Criterion: The switches should allow traffic to go through them so computers in the subnet can receive and send traffic.
    - Priority: 33
    - Dependencies: 24
- Teaches students how to secure and harden the sales subnet switches:
    - ID: 35
    - Type: Security
    - Description: The students will secure the switches inside the sales subnet by disabling unused ports and connections, enabling firewalls, and creating strong credentials to the device.
    - Rationale: Switches are important parts to networks because they connect everything together. This also makes them a target because if an attacker gains access to a switch they gain a good amount of access to the network.
    - Criterion: Any unused ports and connections should be disabled, strong credentials should be used for the device, firewalls should be enabled.
    - Priority: 35
    - Dependencies: 34
- Teaches the students how to setup and configure 24 computers inside the sales subnet:
    - ID: 36

    - Type: Functional
    - Description: Students will assign IP addresses to the computers using the DHCP server and will connect the computers to the sales subnet switches using copper cross-over cables.
    - Rationale: Employees will be using these computers to do work in the company
    - Criterion: All computers should have access to the rest of the network and should have IP addresses assigned to them.
    - Priority: 36
    - Dependencies: 34
- Teaches students how to secure and harden computers inside the sales subnett:
    - ID: 37
    - Type: Security
    - Description: Students will harden computers by enabling firewalls, making strong password policies, and making the correct group policies.
    - Rationale: Limiting employees' access to the network just enough to allow them to work but not enough to impede their production is a good way of securing a network. Employees will be using these computers all the time so it is important they are secure and that attackers can’t use the computers to gain access to the rest of the network.
    - Criterion: All computers should have the correct group policies assigned, strong passwords, and firewalls enabled.
    - Priority: 37
    - Dependencies: 36

Deprecated Functions

- Teaches students how to set up the firewall between the wireless access and main switch in the main building:
    - ID: 21
    - Type: Security
    - Description: The students will learn how to set up a firewall between the main switch in the main building between the wireless access point connected by two copper cross-through cables. The firewall will be set up with rules and ACLs.
    - Rationale: The firewall is necessary so that outside devices connecting to the wireless access point can't harm the network.
    - Criterion: The firewall should have good ACLs and Firewall Rules
    - Priority: 21
    - Dependencies: 5
    - Reason Deprecated: Installing firewall does not work because the wireless access point is not its own network.
- There will be checklist that the lab automatically checks the students network with:
    - ID: 43
    - Type: Security
    - Description: CISCO Packet tracer allows for a checklist to be made that will check and see what the students have implemented within their network to see if they did everything required.

    - Rationale: This will ensure the students do the lab correctly and fully and give the teacher a way to check that the student has done the lab correctly. It will also grade the lab itself.
    - Criterion: Every important detail that my network has should be on the checklist.
    - Priority: 43
    - Dependencies: 38
    - Reason Deprecated: Couldn't figure out how to make the checklist in Cisco Packet Tracer
- The lab will have a script for the students to use when setting up computers:
    - ID: 42
    - Type: Usability
    - Description: The lab will contain scripts for every computer for the different subnets inside the network.
    - Rationale: Scripts lighten the workload on the students learning and reduces the repetitiveness of configuring and securing hundreds of computers within the network. Giving scripts to the students can promote engagement and efficiency and can also be a learning opportunity for them.
    - Criterion: There should be a script for computers within the Sales, Marketing, IT, and Financing subnets respectively that the students can use in the lab. The lab should also describe what the script does.
    - Priority: 42
    - Dependencies: 38
    - Reason Deprecated: Cisco Packet Tracer does not allow for scripts to be used